

Dynamic System Emulation: Fixed Wing Dynamics on a Multicopter

Abdelhakim Amer¹ and Andriy Sarabakha¹

Abstract—This work presents a control framework that enables a multicopter equipped with a camera gimbal to emulate the flight dynamics of a fixed-wing aircraft. The goal is to provide an operationally simple platform for training and simulation that avoids the operating constraints of fixed-wing vehicles, such as minimum airspeed and nonholonomic constraints. A state-input mapping between the two platforms is derived using dynamic feedback linearization. Because the exact inverse is invertible, the actuator and gimbal limits are pulled back through it to characterize, constructively, the set of feasible reference trajectories; a reference governor keeps the emulation within this set, and a singularity-robust formulation bounds the inverse through inverted flight. The framework is evaluated on a three-axis-gimbal platform across a suite of aerobatic manoeuvres, reproducing the fixed-wing camera pose with sub-degree orientation error and sub-metre position error. While evaluations are conducted in simulation, the approach establishes a practical path toward hardware deployment for pilot training, autonomy research, and controller benchmarking.

I. Introduction

Reproducing the behaviour of one dynamical system with another, here called emulation, supports training, algorithm validation, and data collection. Software simulators reproduce a system and its sensory environment in a virtual world [1], [2], but models tuned on synthetic data lose accuracy when transferred to hardware, the so-called sim-to-real gap [3], [4]. Pilot-in-the-loop simulators add a physical motion platform for human training [5], [6], [7], yet their sensory stream is rendered and does not close the loop around the controlled system. This paper considers a third setting, in which one physical system reproduces, in real time, the motion and the camera viewpoint of a second physical system.

Concretely, a multicopter carrying a camera on a gimbal is required to reproduce the camera pose, that is, the position and orientation, of a fixed-wing aircraft. The fixed-wing is constrained by a minimum airspeed and by nonholonomic motion, whereas the multicopter can hover, translate laterally, and fly aggressive manoeuvres, which makes it an inexpensive surrogate for testing perception and control algorithms in real environments [8]. The two platforms are dynamically dissimilar: the fixed-wing produces lift through forward flight and has no static counterpart to the multicopter thrust vectoring [9], [10]. Emulation is therefore a nonlinear output-matching

problem on $SE(3)$ between two underactuated systems, and it raises two questions: whether exact matching is achievable, and by which control law.

Feasibility has been studied through similarity measures, which map the trajectories of one system to those of another and score the mismatch by a cost functional [11], and through lifting techniques, which embed dissimilar dynamics in a common linear space [12], [13]. For the control law, model predictive control enforces input constraints explicitly [14], [15], [16] at a cost that is high for the dimension and bandwidth required here. Exact feedback linearization is an alternative [17], [18]: a regular state feedback transforms the input-output map into a chain of integrators, after which tracking reduces to a linear design. The camera-pose output is not linearizable by static feedback, since the thrust acts only on the translational channel and the decoupling matrix is singular; extending the input with an integrator on the thrust restores a well-defined vector relative degree [19], [20] and yields dynamic feedback linearization. The construction is related to differential flatness [21], used for aggressive quadrotor tracking [22], [23], [24], and to nonlinear output regulation [25].

Two points are usually left open. First, exact linearization is derived without actuator limits, so the synthesized inputs need not be realizable; this is commonly settled by assuming sufficient control authority, which cannot be verified a priori. Second, a two-axis gimbal carries a roll singularity that obstructs full-attitude reproduction. We treat both. The decoupling matrix is invertible by construction, so the actuator and gimbal limits are transported through the inverse to obtain a constructive description of the references a given platform can reproduce, and a third gimbal axis removes the roll singularity.

The contributions are the following. First, we cast physical-to-physical emulation as camera-pose mimicry on $SE(3)$ and show that a multicopter with a three-axis gimbal is output fully actuated and small-time locally controllable for the camera pose. Second, we obtain an exact dynamic-feedback-linearizing law for the mimicry problem and realize it with a single-parameter critically damped gain assignment and a singularity-robust inverse. Third, we characterize the feasible reference set as the preimage of the input-constraint set under the system inverse, and we add a reference governor that confines any reference to this set, turning the customary authority assumption into a checkable condition. Fourth, we validate the framework in simulation over a range of aerobatic manoeuvres. An open-source implementation

This research was supported by the Aarhus University Research Foundation.

¹Abdelhakim Amer and Andriy Sarabakha are with the Department of Electrical and Computer Engineering, Aarhus University, Denmark. {abdelhakim, andriy}@ece.au.dk

accompanies the paper.¹

The paper is organized as follows. Section II states the fixed-wing and multicopter models. Section III establishes the controllability and mimicry results on $SE(3)$. Section IV develops the dynamic-feedback-linearizing controller, the gain assignment, and the feasible set. Section V reports the validation, and Section VI concludes.

II. Preliminaries

Quaternions $\mathbf{q} \in \mathbb{H}_1$ are scalar-first, their conjugate are $\bar{\mathbf{q}} \in \mathbb{H}_1$, and $\mathbf{R}(\mathbf{q}) \in SO(3)$ denotes the corresponding rotation matrix. Also, let $\mathcal{F}_W$ be the inertial world frame.

A. Fixed-Wing Aircraft Dynamics

Fixed-wing aircraft (in Fig. 1) are underactuated dynamic systems. Let $\mathcal{F}_A$ be the aircraft's body frame. The aircraft's pose is given by its position $\mathbf{p}_A = [x_A \ y_A \ z_A]^\top \in \mathbb{R}^3$ in $\mathcal{F}_W$ and a quaternion $\mathbf{q}_A \in \mathbb{H}_1$ mapping $\mathcal{F}_A \rightarrow \mathcal{F}_W$ with ϕ_A , θ_A and ψ_A being roll, pitch and yaw angles, respectively. Aircraft's linear and angular velocities in $\mathcal{F}_A$ are $\mathbf{v}_A \in \mathbb{R}^3$ and $\boldsymbol{\omega}_A \in \mathbb{R}^3$, respectively. The aircraft's state is

$$\mathbf{x}_A = [\mathbf{p}_A \ \mathbf{q}_A \ \mathbf{v}_A \ \boldsymbol{\omega}_A]^\top, \quad (1)$$

while the output is

$$\mathbf{y}_A = [\mathbf{p}_A^\top \ \mathbf{q}_A^\top]^\top. \quad (2)$$

Commonly, the control input to a fixed-wing aircraft is

$$\mathbf{u}_A = [F \ \delta_a \ \delta_e \ \delta_r]^\top, \quad (3)$$

where F is the thrust generated by the engines, δ_a , δ_e , and δ_r are aileron, elevator, and rudder deflections. Aerodynamic forces acting on a fixed-wing aircraft defined in the wind frame $\mathcal{F}_V$ are $\mathbf{f}_{\text{aero}} = [-D \ Y \ L]^\top$, with $D \propto v^2$ is drag, Y is side force, $L \propto v^2$ is lift, and v is airspeed. The total force acting on the fixed-wing aircraft in $\mathcal{F}_A$ is $\mathbf{f}_A = \mathbf{R}(\alpha)\mathbf{f}_{\text{aero}} + [F \ 0 \ 0]^\top + \mathbf{R}(\mathbf{q}_A)^\top [0 \ 0 \ -g]^\top$,

¹Repository: https://github.com/abdelhakim96/Ghost_Tracking_DFL

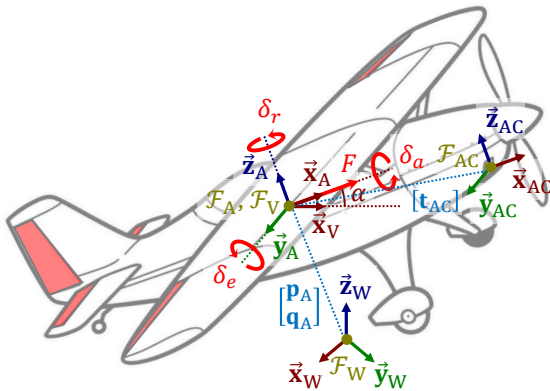


Fig. 1. Fixed-wing aircraft shown with its reference frames, control inputs (in red) and output variables (in blue).

where α is the angle of attack, and g is the gravitational constant. The total torque acting on the fixed-wing aircraft in $\mathcal{F}_A$ is $\boldsymbol{\tau}_A = \mathbf{B}_\tau [\delta_a \ \delta_e \ \delta_r]^\top$, where $\mathbf{B}_\tau$ is the control-effectiveness map. Using the Newton-Euler equations, the kinematics and dynamics of a fixed-wing aircraft are:

$$f_A : \begin{cases} \dot{\mathbf{p}}_A = \mathbf{R}(\mathbf{q}_A) \mathbf{v}_A \\ \dot{\mathbf{q}}_A = \frac{1}{2} \mathbf{G}(\mathbf{q}_A) \boldsymbol{\omega}_A \\ \dot{\mathbf{v}}_A = -\boldsymbol{\omega}_A \times \mathbf{v}_A + \frac{1}{m_A} \mathbf{f}_A \\ \dot{\boldsymbol{\omega}}_A = -(\mathbf{J}_A)^{-1} \boldsymbol{\omega}_A \times (\mathbf{J}_A \boldsymbol{\omega}_A) + (\mathbf{J}_A)^{-1} \boldsymbol{\tau}_A \end{cases}, \quad (4)$$

in which $\mathbf{G}(\mathbf{q}) \in \mathbb{R}^{4 \times 3}$ is the quaternion kinematics matrix, m_A is aircraft's mass, and $\mathbf{J}_A \in \mathbb{R}^{3 \times 3}$ is matrix of inertia.

B. Multicopter with 3-Axis Gimbal Dynamics

Multicopters (in Fig. 2) are also underactuated dynamic systems. Let $\mathcal{F}_M$ be the multicopter's body frame. The multicopter's pose is given by its position $\mathbf{p}_M = [x_M \ y_M \ z_M]^\top \in \mathbb{R}^3$ in $\mathcal{F}_W$ and a quaternion $\mathbf{q}_M \in \mathbb{H}_1$ mapping $\mathcal{F}_M \rightarrow \mathcal{F}_W$ with ϕ_M , θ_M and ψ_M being roll, pitch and yaw angles, respectively. Multicopter's linear and angular velocities in $\mathcal{F}_M$ are $\mathbf{v}_M \in \mathbb{R}^3$ and $\boldsymbol{\omega}_M \in \mathbb{R}^3$, respectively. The multicopter is equipped with a camera mounted on a 3-axis gimbal with joint angles $\boldsymbol{\Theta}_G = [\phi_G \ \theta_G \ \psi_G]^\top$ and quaternion $\mathbf{q}_G = \mathbf{q}_x(\phi_G) \otimes \mathbf{q}_y(\theta_G) \otimes \mathbf{q}_z(\psi_G) \in \mathbb{H}_1$, which maps from the multicopter's camera frame $\mathcal{F}_{MC}$ to $\mathcal{F}_M$. Let the camera lever arm in $\mathcal{F}_M$ be $\mathbf{t}_G \in \mathbb{R}^3$. The state of a multicopter with a 3-axis gimbal is

$$\mathbf{x}_M = [\mathbf{p}_M \ \mathbf{q}_M \ \mathbf{v}_M \ \boldsymbol{\omega}_M \ \boldsymbol{\Theta}_G]^\top, \quad (5)$$

while the output is the camera's pose:

$$\mathbf{y}_{MC} = [\mathbf{p}_M^\top + (R(\mathbf{q}_M) \mathbf{t}_G)^\top \ (\mathbf{q}_M \otimes \mathbf{q}_G)^\top]^\top. \quad (6)$$

Commonly, the control input to a multicopter is

$$\mathbf{u}_M = [T \ \tau_\phi \ \tau_\theta \ \tau_\psi]^\top, \quad (7)$$

where T is total thrust, τ_ϕ , τ_θ , and τ_ψ are torques around x -, y - and z -axes, respectively. Control inputs to the

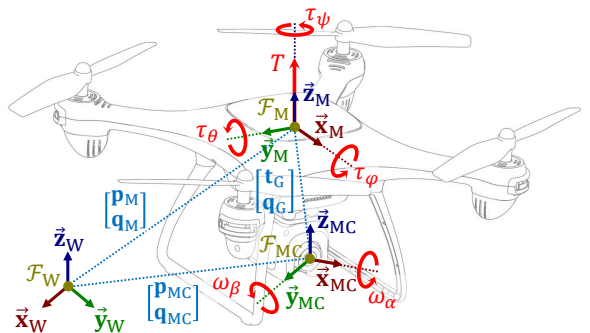


Fig. 2. Multicopter with a camera gimbal shown with its reference frames, control inputs (in red) and output variables (in blue).

gimbal are

$$\mathbf{u}_G = \dot{\boldsymbol{\Theta}}_G = [\dot{\phi}_G \quad \dot{\theta}_G \quad \dot{\psi}_G]^\top, \quad (8)$$

i.e. the gimbal roll, pitch and yaw rates. The total force acting on the multicopter in $\mathcal{F}_M$ is $\mathbf{f}_M = [0 \quad 0 \quad T]^\top + \mathbf{R}(\mathbf{q}_M)^\top [0 \quad 0 \quad -g]^\top$; while the total torque in $\mathcal{F}_M$ is $\boldsymbol{\tau}_M = [\tau_\phi \quad \tau_\theta \quad \tau_\psi]^\top$. So, the kinematics and dynamics of a multicopter with a gimbal are:

$$f_M : \begin{cases} \dot{\mathbf{p}}_M = \mathbf{R}(\mathbf{q}_M) \mathbf{v}_M \\ \dot{\mathbf{q}}_M = \frac{1}{2} \mathbf{G}(\mathbf{q}_M) \boldsymbol{\omega}_M \\ \dot{\mathbf{v}}_M = -\boldsymbol{\omega}_M \times \mathbf{v}_M + \frac{1}{m_M} \mathbf{f}_M \\ \dot{\boldsymbol{\omega}}_M = -(\mathbf{J}_M)^{-1} \boldsymbol{\omega}_M \times (\mathbf{J}_M \boldsymbol{\omega}_M) + (\mathbf{J}_M)^{-1} \boldsymbol{\tau}_M \\ \dot{\boldsymbol{\Theta}}_G = \mathbf{u}_G \end{cases}, \quad (9)$$

in which m_M is the mass of multicopter with gimbal, and $\mathbf{J}_M \in \mathbb{R}^{3 \times 3}$ is matrix of inertia.

III. Camera-Pose Controllability and Mimicry on $SE(3)$

Assume the fixed-wing carries a camera that is body-aligned and located at a fixed lever arm $\mathbf{t}_{AC} \in \mathbb{R}^3$ expressed in $\mathcal{F}_A$. Then, the fixed-wing camera pose is

$$\mathbf{y}_{AC} = [\mathbf{p}_{AC}^\top \quad \mathbf{q}_{AC}^\top] = [(\mathbf{p}_A + \mathbf{R}(\mathbf{q}_A) \mathbf{t}_{AC})^\top \quad \mathbf{q}_A^\top]. \quad (10)$$

We expose (i) a translational “virtual” input $\mathbf{a}_{MC} \in \mathbb{R}^3$ realized by the standard $SE(3)$ inner loop (third line of (9)), and (ii) the gimbal rate inputs $\mathbf{u}_G = \dot{\boldsymbol{\Theta}}_G \in \mathbb{R}^3$. The multicopter body attitude $\mathbf{q}_M$ is dictated by the thrust direction required for translation, with its free yaw held at zero; the camera orientation is therefore matched by the three gimbal degrees of freedom.

From (6), the position of the multicopter’s camera in the $\mathcal{F}_W$ is

$$\mathbf{p}_{MC} = \mathbf{p}_M + \mathbf{R}(\mathbf{q}_M) \mathbf{t}_G, \quad (11)$$

while the rotation is

$$\mathbf{q}_{MC} = \mathbf{q}_M \otimes \mathbf{q}_G. \quad (12)$$

Then, by (11)

$$\ddot{\mathbf{p}}_{MC} = \mathbf{a}_M + \boldsymbol{\Delta}(\mathbf{p}_{MC}, \mathbf{q}_M, \boldsymbol{\omega}_M, \dot{\boldsymbol{\omega}}_M), \quad (13)$$

where $\boldsymbol{\Delta}(\mathbf{p}_{MC}, \mathbf{q}_M, \boldsymbol{\omega}_M, \dot{\boldsymbol{\omega}}_M)$ collects the known lever-arm coupling terms, and by (12)

$$\dot{\mathbf{q}}_{MC} = \frac{1}{2} \mathbf{G}(\mathbf{q}_{MC}) \boldsymbol{\omega}_{MC}, \quad (14)$$

where the camera angular velocity decomposes as $\boldsymbol{\omega}_{MC} = \mathbf{R}(\mathbf{q}_G)^\top \boldsymbol{\omega}_M + \mathbf{J}_G(\phi_G, \theta_G) \mathbf{u}_G$ with the 3-axis gimbal Jacobian

$$\mathbf{J}_G(\phi, \theta) = \begin{bmatrix} 1 & 0 & \sin \theta \\ 0 & \cos \phi & -\sin \phi \cos \theta \\ 0 & \sin \phi & \cos \phi \cos \theta \end{bmatrix}, \quad (15)$$

whose determinant is

$$\det \mathbf{J}_G(\phi, \theta) = \cos \theta. \quad (16)$$

Remark 1: A 2-axis (roll–pitch) gimbal combined with the multicopter’s free yaw has input-twist determinant $\cos \phi_G \cos \theta_G$, hence a codimension-one singular set $\cos \phi_G \cos \theta_G = 0$ that obstructs full-roll maneuvers. The third gimbal axis removes the roll singularity: by (16) the only singularity is the gimbal pitch $\theta_G = \pm \frac{\pi}{2}$.

Assumption 1 (Non-Singularity): The gimbal pitch stays away from the singular set: $\theta_G \neq \pm \frac{\pi}{2}$.

Remark 2: This assumption can be relaxed by a planner that keeps the camera-pitch reference clear of the singular set.

A. Output Full Actuation & STLC on $SE(3)$

Consider the block-triangular Jacobian from inputs to camera acceleration and twist:

$$\begin{bmatrix} \ddot{\mathbf{p}}_{MC} \\ \boldsymbol{\omega}_{MC} \end{bmatrix} = \underbrace{\begin{bmatrix} \mathbf{I}_3 & \mathbf{0} \\ \mathbf{0} & \mathbf{J}_G(\phi_G, \theta_G) \end{bmatrix}}_{\mathbf{J}} \begin{bmatrix} \mathbf{a}_{MC} \\ \mathbf{u}_G \end{bmatrix} + \begin{bmatrix} \boldsymbol{\Delta} \\ \mathbf{R}(\mathbf{q}_G)^\top \boldsymbol{\omega}_M \end{bmatrix}. \quad (17)$$

a) Translational rank: Thrust vectoring yields $\mathbf{a}_{MC} = -g\mathbf{e}_3 + \frac{T}{m_M} \mathbf{b}_3$; variations $\delta \mathbf{a}_{MC} = \frac{\delta T}{m_M} \mathbf{b}_3 + \frac{T}{m_M} \delta \mathbf{b}_3$ span $\mathbb{R}^3$. Hence, $\mathbf{a}_{MC} \mapsto \ddot{\mathbf{p}}_{MC}$ is locally surjective.

b) Rotational rank: By (16), $\text{rank}(\mathbf{J}_G) = 3$ whenever $\cos \theta_G \neq 0$, so the gimbal rates span the camera angular velocity.

Therefore, $\text{rank}(\mathbf{J}) = 6$ for all $\theta_G \neq \pm \frac{\pi}{2}$, and the camera pose is output fully actuated and small-time locally controllable (STLC) on $SE(3)$.

B. Camera-Pose Mimicry

Now we show that multicopter and gimbal control inputs exist so that

$$\mathbf{y}_{MC}(t) \equiv \mathbf{y}_{AC}(t) \quad \forall t. \quad (18)$$

Assumption 2 (Authority): Thrust and torque of the multicopter, as well as gimbal angle/rate limits, are not violated along the trajectories of interest.

a) Position matching: Equality of camera positions, $\mathbf{p}_M + \mathbf{R}(\mathbf{q}_M) \mathbf{t}_G = \mathbf{p}_A + \mathbf{R}(\mathbf{q}_A) \mathbf{t}_{AC}$, is realised by commanding the multicopter position to the lever-arm-compensated target

$$\mathbf{p}_M = \mathbf{p}_A + \mathbf{R}(\mathbf{q}_A) \mathbf{t}_{AC} - \mathbf{R}(\mathbf{q}_M) \mathbf{t}_G. \quad (19)$$

Taking two derivatives gives the desired multicopter’s acceleration

$$\mathbf{a}_M = \ddot{\mathbf{p}}_A + \frac{d^2 \mathbf{R}(\mathbf{q}_A)}{dt^2} \mathbf{t}_{AC} - \frac{d^2 \mathbf{R}(\mathbf{q}_M)}{dt^2} \mathbf{t}_G, \quad (20)$$

which the $SE(3)$ inner loop (third line of (9)) can track so that $\mathbf{p}_M + \mathbf{R}(\mathbf{q}_M) \mathbf{t}_G \rightarrow \mathbf{p}_A + \mathbf{R}(\mathbf{q}_A) \mathbf{t}_{AC}$.

b) Orientation matching: We require $\mathbf{q}_M \otimes \mathbf{q}_G = \mathbf{q}_A$. The multicopter attitude is dictated by translation,

$$\mathbf{q}_M = \mathbf{q}_z(\psi_M) \otimes \mathbf{q}_{\text{tilt}}, \quad (21)$$

where $\mathbf{q}_{\text{tilt}}$ sets the thrust direction required by $\mathbf{a}_M$ and the free yaw is held at $\psi_M = 0$. The residual rotation is assigned to the gimbal,

$$\mathbf{q}_{\text{rel}} = \bar{\mathbf{q}}_M \otimes \mathbf{q}_A. \quad (22)$$

Because the 3-axis gimbal parameterizes the full rotation group, there exist $(\phi_G, \theta_G, \psi_G)$ with $\theta_G \neq \pm \frac{\pi}{2}$ such that

$$\mathbf{q}_G = \mathbf{q}_x(\phi_G) \otimes \mathbf{q}_y(\theta_G) \otimes \mathbf{q}_z(\psi_G) = \mathbf{q}_{\text{rel}}. \quad (23)$$

This yields $\mathbf{q}_M \otimes \mathbf{q}_G = \mathbf{q}_A$, so the camera orientations match exactly. The third gimbal axis is what makes this hold for arbitrary $\mathbf{q}_{\text{rel}}$, in particular through full rolls and inverted attitudes, whereas a 2-axis gimbal realizes only the roll-pitch submanifold.

Under Assumptions 1 and 2, the multicopter with a 3-axis gimbal is output fully actuated and STLC on $SE(3)$ for the camera pose, and control inputs exist which realise $\mathbf{y}_{MC}(t) \equiv \mathbf{y}_{AC}(t)$ for any feasible fixed-wing camera trajectory $\mathbf{y}_{AC} \in \mathcal{F}$ (the feasible set is characterized in Section IV-B). With identical intrinsics, the rendered image streams are indistinguishable to the viewer.

IV. Proposed Method

Given the fixed-wing camera trajectory $\mathbf{y}_{AC}(t)$ in (10), we want to compute multicopter and gimbal inputs,

$$\mathbf{u}_{MC} = \begin{bmatrix} \mathbf{u}_M \\ \mathbf{u}_G \end{bmatrix}, \quad (24)$$

such that the multicopter camera pose $\mathbf{y}_{MC}(t)$ in (6) satisfies $\mathbf{y}_{MC}(t) \equiv \mathbf{y}_{AC}(t) \quad \forall t$.

To enable multicopter tracking aggressive manoeuvres, non-linear control techniques must be used. Dynamic feedback linearization (DFL) could be designed to achieve better convergence and robustness. Through a change of variables, DFL transforms a nonlinear system into an equivalent linear, controllable and observable one. Given a non-linear system

$$\begin{cases} \dot{\mathbf{x}} = f(\mathbf{x}) + g(\mathbf{x})\mathbf{u} \\ \mathbf{y} = h(\mathbf{x}), \end{cases} \quad (25)$$

where $\mathbf{x}$ is the system state, $\mathbf{u}$ is the system input, $\mathbf{y}$ is the system output, $f(\mathbf{x})$, $g(\mathbf{x})$ and $h(\mathbf{x})$ are vector fields in $\mathbb{R}^n$. The task of feedback linearization is to identify a static state feedback control law of the following form:

$$\mathbf{u} = \alpha(\mathbf{x}) + \beta(\mathbf{x})\mathbf{v}, \quad (26)$$

where $\mathbf{v}$ is a new control input, $\alpha(\mathbf{x})$ and $\beta(\mathbf{x})$ are smooth functions defined in a neighbourhood of some point $\mathbf{x}_0 \in \mathbb{R}^n$ and $\beta(\mathbf{x}_0) \neq 0$, such that the closed-loop system composed of (25) and (26) behaves as a linear and completely accessible system. For system (25), let N_I be the number of inputs, N_O be the number of outputs, $\mathbf{r} = [r_1 \ \cdots \ r_{N_O}]$ be the vector relative degree. Then, let's define

$$\Delta(\mathbf{x}) = \begin{bmatrix} L_{g_1} L_f^{r_1-1} h_1(\mathbf{x}) & \cdots & L_{g_{N_I}} L_f^{r_1-1} h_1(\mathbf{x}) \\ \vdots & \ddots & \vdots \\ L_{g_1} L_f^{r_{N_O}-1} h_{N_O}(\mathbf{x}) & \cdots & L_{g_{N_I}} L_f^{r_{N_O}-1} h_{N_O}(\mathbf{x}) \end{bmatrix} \quad (27)$$

and

$$\mathbf{b}(\mathbf{x}) = \begin{bmatrix} L_f^{r_1} h_1(\mathbf{x}) & \cdots & L_f^{r_{N_O}} h_{N_O}(\mathbf{x}) \end{bmatrix}^T, \quad (28)$$

where $L_f^r h$ denotes the r -th Lie derivative of the function h with respect to the vector field f . The main result about the input-output decoupling problem is that this problem is solvable if and only if the matrix $\Delta(\mathbf{x})$ is non-singular. In that case, the static state feedback (26) with

$$\begin{cases} \alpha(\mathbf{x}) = -(\Delta(\mathbf{x}))^{-1} \mathbf{b}(\mathbf{x}) \\ \beta(\mathbf{x}) = (\Delta(\mathbf{x}))^{-1} \end{cases} \quad (29)$$

renders the closed-loop system linear and decoupled from an input-output point of view.

To feedback linearize a system, the necessary and sufficient condition for the solvability of the state space exact linearization problem is [19]:

$$\sum_{i=1}^{N_O} r_i = N, \quad (30)$$

where N is the system's rank.

For the system (9) with the seven outputs $\mathbf{y}_O = [\mathbf{p}_{MC}^\top \ q_3 \ \boldsymbol{\Theta}_G^\top]^\top$ (camera position, multicopter yaw component q_3 , and the three gimbal angles), the vector relative degree is $\mathbf{r} = [2 \ 2 \ 2 \ 2 \ 1 \ 1 \ 1]$ with $N = 15$ [26]. Since $\sum_i r_i = 11 \neq 15$, the condition (30) is not satisfied. Therefore, the input-output decoupling problem is not solvable for the system (9) by means of a static state feedback control law. In fact, $\Delta(\mathbf{x}_M)$ is singular for any $\mathbf{x}_M$ and is not invertible. So, the system (9) can not be transformed into an equivalent linear and controllable one by static state feedback. This is due to the fact that $\mathbf{a}_M$ is affected only by thrust T and not by other control inputs. In order to get $\Delta(\mathbf{x})$ non-singular, the appearance of T should be delayed to higher order derivatives of $\mathbf{a}_M$. This can be achieved by a dynamic compensator which incorporates an additional set of state variables $\boldsymbol{\chi}$ to achieve higher relative degrees [20]. By setting T equal to the output of an auxiliary dynamic system driven by a new reference input $\hat{T}$. The simplest way in which this result can be achieved is to set this auxiliary dynamic system equal to a double integrator. In this case, $\boldsymbol{\chi} = [\chi_1 \ \chi_2]^\top$. So, T is set equal to the output of a double integrator driven by $\hat{T}$. The other input variables in (24) are unchanged. Now, T is not

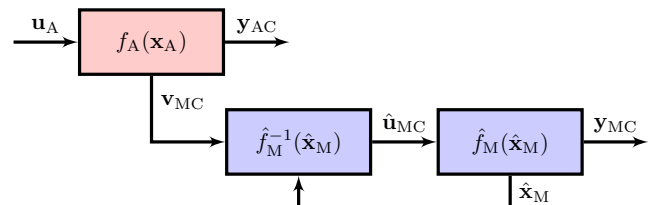


Fig. 3. Overview of the proposed framework.

any-more an input for the system (9), but becomes the internal state for the new dynamical system

$$\hat{f}_M : \begin{cases} \dot{\mathbf{p}}_M = \mathbf{R}(\mathbf{q}_M) \mathbf{v}_M \\ \dot{\mathbf{q}}_M = \frac{1}{2} \mathbf{G}(\mathbf{q}_M) \boldsymbol{\omega}_M \\ \dot{\mathbf{v}}_M = -\boldsymbol{\omega}_M \times \mathbf{v}_M + \frac{1}{m_M} \begin{bmatrix} 0 & 0 & \chi \end{bmatrix}^\top \\ \dot{\boldsymbol{\omega}}_M = -(\mathbf{J}_M)^{-1} \boldsymbol{\omega}_M \times (\mathbf{J}_M \boldsymbol{\omega}_M) + (\mathbf{J}_M)^{-1} \boldsymbol{\tau}_M \\ \dot{\boldsymbol{\theta}}_G = \mathbf{u}_G \\ \ddot{\chi} = \ddot{T} \end{cases} \quad (31)$$

The new control inputs to the system (31) are

$$\hat{\mathbf{u}}_{MC} = [\ddot{T} \quad \tau_\phi \quad \tau_\theta \quad \tau_\psi \quad \dot{\phi}_G \quad \dot{\theta}_G \quad \dot{\psi}_G]^\top. \quad (32)$$

The input-output decoupling problem is solvable for the system (9) by means of a dynamic feedback control law, if it is solvable via static feedback for the extended system (31). Since the extended system has $\hat{\mathbf{r}} = [4 \ 4 \ 4 \ 2 \ 1 \ 1 \ 1]^\top$ and $\hat{N} = 17$, with $\sum_i \hat{r}_i = 17 = \hat{N}$, the condition (30) is satisfied. Thus, the input-output decoupling problem is solvable for the system (31) by means of a dynamic feedback control law as in (26):

$$\hat{\mathbf{u}}_{MC} = \alpha(\hat{\mathbf{x}}_{MC}) + \beta(\hat{\mathbf{x}}_{MC}) \mathbf{v}_{MC}, \quad (33)$$

where $\alpha(\hat{\mathbf{x}}_{MC})$ and $\beta(\hat{\mathbf{x}}_{MC})$ are computed using (29), and

$$\mathbf{v}_{MC} = [\mathbf{p}_{AC}^\top \quad \dot{\mathbf{p}}_{AC}^\top \quad \ddot{\mathbf{p}}_{AC}^\top \quad \ddot{\mathbf{p}}_{AC}^\top \quad \ddot{\mathbf{p}}_{AC}^\top \quad \mathbf{q}_A^\top \quad \dot{\mathbf{q}}_A^\top \quad \ddot{\mathbf{q}}_A^\top]^\top. \quad (34)$$

An overview of the proposed framework is shown in Fig. 3.

A. Closed-Loop Realization

The feedback (33) renders each output an integrator chain, $y_i^{(r_i)} = v_i$, decoupled from the others. We close the chains with a two-degree-of-freedom law

$$v_i = y_{\text{ref},i}^{(r_i)} + \sum_{j=0}^{r_i-1} c_{i,j} (y_{\text{ref},i}^{(j)} - y_i^{(j)}), \quad (35)$$

so the tracking error $e_i = y_i - y_{\text{ref},i}$ obeys the certified linear dynamics $e_i^{(r_i)} + \sum_j c_{i,j} e_i^{(j)} = 0$.

Remark 3 (One-parameter gains): Placing all poles of every error chain at a single bandwidth $-\omega$ forces the gains to the binomial coefficients $c_{i,j} = \binom{r_i}{j} \omega^{r_i-j}$. The closed loop is then critically damped (monotone error decay, no overshoot) and tuned by the single parameter ω , removing any hand-chosen gain vector.

Remark 4 (Singularity-robust inverse): The decoupling matrix $\Delta(\hat{\mathbf{x}}_{MC})$ carries a $1/q_0$ factor that diverges as the multicopter inverts ($q_0 \rightarrow 0$). Evaluating $\beta = (\Delta)^{-1}$ with $|q_0|$ floored at a small ε yields a damped-inverse law that is exact away from the singular set and bounded through inverted flight, at the cost of an $O(\varepsilon)$ deviation in a neighbourhood of $q_0 = 0$.

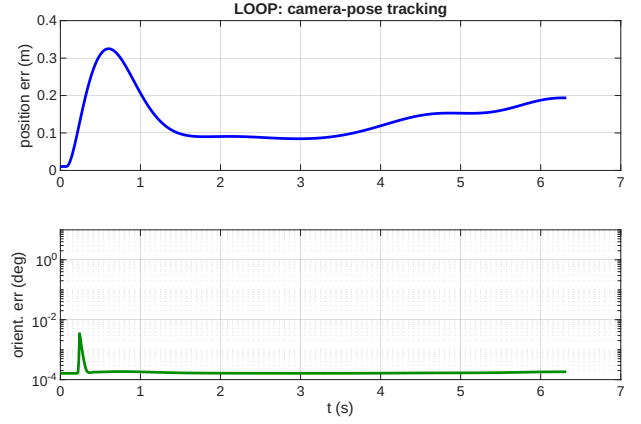


Fig. 4. Camera-pose tracking for the loop: position error (top) stays sub-metre while the camera orientation error (bottom, log scale) remains below 0.01° throughout, despite the airframe inverting.

B. Feasible Set and Reference Governor

The exact inverse maps a reference and its derivatives to the required inputs. Collecting the actuator and gimbal limits into $\mathcal{U} = \{0 \leq T \leq T_{\max}, \|\boldsymbol{\tau}_M\| \leq \tau_{\max}, \|\mathbf{u}_G\| \leq \dot{\theta}_{\max}, \cos \theta_G \neq 0\}$, the nonsingularity of β pulls these limits back exactly into the linearizing coordinates.

Lemma 1 (Feasible set): Let $\mathbf{v}_{MC}(t)$ collect the reference $\mathbf{y}_{AC}(t)$ and its derivatives as in (33), and define

$$\mathcal{F} = \{\mathbf{y}_{AC}(\cdot) : \alpha(\hat{\mathbf{x}}_{MC}(t)) + \beta(\hat{\mathbf{x}}_{MC}(t)) \mathbf{v}_{MC}(t) \in \mathcal{U} \ \forall t\}. \quad (36)$$

Then for every $\mathbf{y}_{AC} \in \mathcal{F}$ the dynamic feedback (33) realizes $\mathbf{y}_{MC}(t) \equiv \mathbf{y}_{AC}(t)$ with inputs that satisfy $\mathcal{U}$.

Lemma 1 replaces the non-constructive Assumption 2 with a checkable condition: the authority requirement is exactly the membership $\mathbf{y}_{AC} \in \mathcal{F}$, the preimage of the input-constraint set $\mathcal{U}$ under the system inverse.

Remark 5 (Reference governor): For a reference outside $\mathcal{F}$, a time-scaling governor $\mathbf{y}_{AC}^\lambda(t) = \mathbf{y}_{AC}(\lambda t)$, $\lambda \in (0, 1]$, scales the k -th reference derivative by λ^k ; the largest λ with $\mathbf{y}_{AC}^\lambda \in \mathcal{F}$ gives the most aggressive feasibly-emulated version of the maneuver. As thrust-to-weight and rate demands are mass-invariant under feedback linearization, $\mathcal{F}$ and λ depend on the reference and the platform limits, not on the vehicle mass.

V. Validation Results

We validate the proposed framework on a representative 5kg multicopter equipped with a 3-axis gimbal, commanded to reproduce the camera pose of a Red Bull Edge 540 [27] aerobatic aircraft. The Edge 540 is chosen for its aggressive maneuvering envelope, providing a demanding reference. Vehicle parameters are listed in Table I. The reference trajectories are generated by the full 6-DOF fixed-wing model (4), which supplies the camera position and its derivatives up to snap, and the camera attitude, as the inputs $\mathbf{v}_{MC}$ to the dynamic-feedback-linearizing law (33).

A. Controller Realization

The exact linearization of Section IV reduces each output to an integrator chain $y_i^{(r_i)} = v_i$. We close these chains with three theory-grounded elements and no ad-hoc clamps:

- One-parameter critically-damped gains. All poles of every error chain are placed at a single bandwidth $-\omega$, so the feedback gains are the binomial coefficients of $(s + \omega)^{r_i}$. There is no hand-tuned gain vector; we use $\omega = 30 \text{ rad s}^{-1}$.
- Singularity-robust inverse. The decoupling map carries a $1/q_0$ term that diverges at the body-attitude representation singularity ($q_0 \rightarrow 0$, i.e. the airframe inverting past 90°). We floor $|q_0|$ when evaluating the inverse, keeping the linearizing law bounded through inverted flight while remaining exact away from the singular set.
- Reference governor. Since the thrust and rate demands of a maneuver are set by the reference and are mass-invariant under feedback linearization, we admit a pace factor $\lambda \in (0, 1]$ that time-scales the reference so the inverse inputs stay within the actuator and gimbal limits, i.e. within the feasible set $\mathcal{F}$.

B. Tracking Performance

We evaluate six aerobatic maneuvers: a vertical loop, an axial roll, a soft roll, a barrel roll, a Split-S, and a Cuban-8. Each reference is paced by the governor to a feasible thrust-to-weight ($T/W \leq 3$). Table II reports, per maneuver, the governor pace λ , the peak thrust-to-weight, the mean-absolute camera-position error, and the maximum camera-orientation (geodesic) error.

Across all maneuvers the camera-orientation error stays below 0.05° and the camera-position error stays sub-metre, confirming high-fidelity reproduction of the fixed-wing camera pose. Fig. 4 shows the loop in detail: both the position error and the (log-scale) orientation error remain small throughout, even as the airframe passes through inverted flight.

Fig. 5 overlays, in three dimensions, the fixed-wing camera path (red) and the multicopter camera path (blue) for four maneuvers, with the (transparent) aircraft drawn at intervals along with the multicopter camera frame (green boresight). The two paths are visually indistinguishable, and the camera boresight stays aligned with

TABLE I
Vehicle Parameters

Fixed-Wing (Red Bull Edge 540)	
Mass	530 kg
Moments of inertia	(2419, 2133, 4552) $\text{kg} \cdot \text{m}^2$
Wingspan	7.40 m
Wing Area	9.10 m^2
Max Speed	118 m/s
Multicopter (3-axis gimbal)	
Mass	5.0 kg
Moments of inertia	(0.15, 0.15, 0.25) $\text{kg} \cdot \text{m}^2$
Arm length	0.35 m

TABLE II
Camera-Pose Tracking (governed, $T/W \leq 3$)

Maneuver	λ	peak T/W	MAE pos. (m)	max orient. (deg)
Loop	0.87	3.0	0.14	0.003
Roll	0.60	3.0	0.07	0.006
Soft roll	0.60	3.0	0.11	0.021
Barrel roll	0.60	3.0	0.10	0.028
Split-S	0.29	3.0	0.06	0.003
Cuban-8	0.35	3.0	0.07	0.008

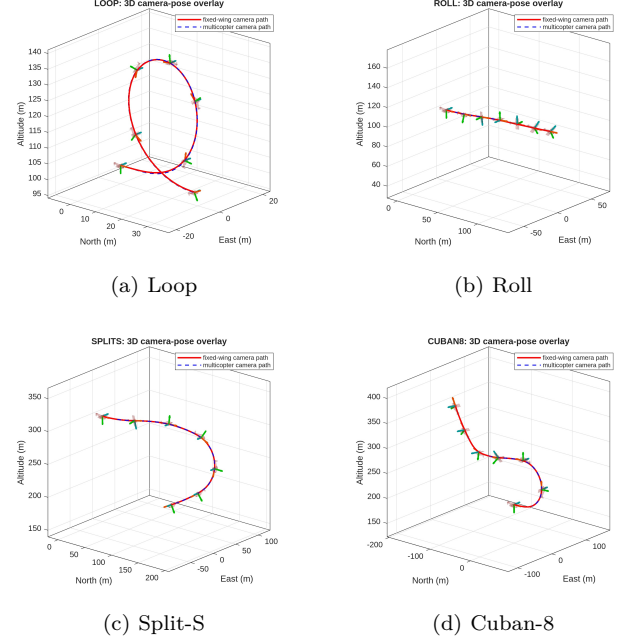


Fig. 5. 3D camera-pose overlays. Fixed-wing camera path (red) and multicopter camera path (blue, dashed) coincide; the transparent aircraft and the multicopter camera frame (green boresight) are shown at intervals. The airframe attitude varies while the camera stays locked on the reference.

the aircraft, confirming that position and orientation are both reproduced. The figure also makes the decoupling explicit: the airframe attitude (transparent aircraft pose) varies strongly along the trajectory while the camera frame remains locked on the reference.

The decoupling is the key behaviour. To match the fixed-wing position, the multicopter must reproduce its specific-force vector; the thrust direction tracks the aircraft's specific force to within $\sim 10^\circ$, so the airframe tilts and even inverts (body tilt reaching 170 – 180°) to vector thrust along the curved, high-load trajectory. The 3-axis gimbal simultaneously absorbs the residual rotation $\mathbf{q}_G = \bar{\mathbf{q}}_M \otimes \mathbf{q}_A$, so the camera orientation stays locked on the fixed-wing's. The airframe thus reproduces the load factor experienced by the fixed-wing while the camera viewpoint remains faithful, which is the goal of physical-to-physical emulation.

C. Control Inputs and Feasibility

Fig. 6 shows the realized control effort for the loop and roll against the platform limits (thrust-to-weight

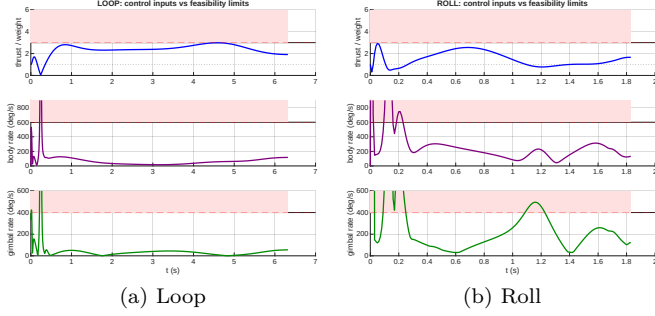


Fig. 6. Control inputs versus feasibility limits (shaded = infeasible). Thrust-to-weight, body rate, and gimbal rate all remain within bounds after the initial transient.

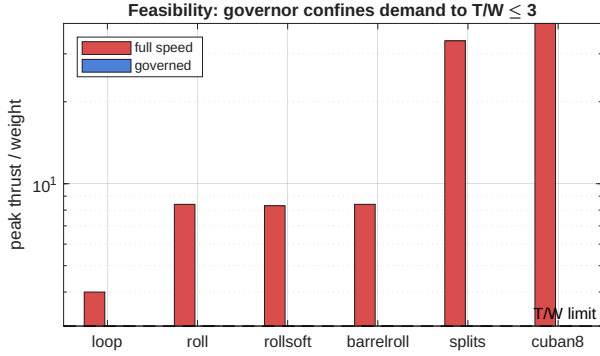


Fig. 7. Peak thrust-to-weight per maneuver at full speed versus governed; the governor confines the demand below the platform limit ($T/W \leq 3$).

≤ 3 , body rate ≤ 600 deg/s, gimbal rate ≤ 400 deg/s). After a brief initial transient, all three inputs remain within their limits for the full maneuver: thrust-to-weight oscillates between roughly 1 and 3 as the aircraft loads and unloads, while the body and gimbal rates settle well below their bounds.

Because the exact inverse reproduces the reference acceleration regardless of vehicle mass, the thrust-to-weight demand is a property of the trajectory, not the platform. At full speed the Edge 540 maneuvers demand $4\text{--}39\times$ thrust-to-weight (Fig. 7), far beyond any realizable multicopter. The reference governor pulls each maneuver into $\mathcal{F}$ (Lemma 1) by pacing, recovering $T/W \leq 3$ while preserving exact tracking (Table II).

Feasibility is bounded jointly by thrust authority and the gimbal kinematic range. Slowing a maneuver reduces the thrust demand but also reduces the airframe tilt, shifting orientation onto the gimbal and toward its $\theta_G = \pm \pi/2$ singularity. The loop and the axial roll satisfy both limits comfortably (gimbal margin $\cos \theta_G \geq 0.38$); the remaining maneuvers track exactly and respect thrust but operate nearer the gimbal limit. This interplay, in which the body flip extends the orientation range beyond the gimbal at the cost of thrust authority, characterizes the boundary of $\mathcal{F}$ for a given platform.

D. Discussion

The results validate the framework of Sections III and IV: the multicopter-gimbal system achieves camera-pose mimicry $\mathbf{y}_{MC}(t) \equiv \mathbf{y}_{AC}(t)$ with sub-degree orientation error and sub-metre position error across a suite of aerobatic maneuvers, using an exact, singularity-robust feedback-linearizing controller with a single stabilization parameter. The reference governor renders any reference feasible by confining it to $\mathcal{F}$ (Lemma 1), replacing the non-constructive Assumption 2 with a checkable condition derived directly from the system inverse.

VI. Conclusions and Future Work

This work has designed and validated a control framework for physical-to-physical emulation, demonstrating that a multicopter equipped with a 3-axis gimbal can dynamically replicate the camera viewpoint and flight dynamics of a fixed-wing aircraft. Our approach, based on exact dynamic feedback linearization, provides a practical method for achieving high-fidelity mimicry of a primary system's key observables, and characterizes constructively the set of references a given platform can emulate. This capability enables algorithm validation and operator training in real-world conditions at a negligible cost and with significantly reduced operational risk compared to deploying the high-value fixed-wing planes. Furthermore, the principles developed can be generalized to other classes of dissimilar dynamic systems, such as ground or underwater vehicles, opening new paradigms for cross-system validation.

Appendix

This appendix derives the dynamic-feedback-linearizing law (33) explicitly, recovering by hand the decoupling matrix and drift that the open-source code generates symbolically. We take the lever arm $\mathbf{t}_G = \mathbf{0}$ (camera at the multicopter centre of mass), so that $\mathbf{p}_{MC} = \mathbf{p}_M$, and write $\mathbf{b}_3 = \mathbf{R}(\mathbf{q}_M)\mathbf{e}_3$ for the body thrust axis and $\boldsymbol{\omega}_w = \mathbf{R}(\mathbf{q}_M)\boldsymbol{\omega}_M$ for the body rate in $\mathcal{F}_W$. The skew operator is $[\cdot]_\times$.

A. Extended control-affine model

After the thrust double integrator $\dot{\zeta} = \xi$, $\dot{\xi} = \ddot{T}$ of Section IV, the extended state is $\hat{\mathbf{x}} = [\mathbf{p}_M^\top, \mathbf{q}_M^\top, \mathbf{v}^\top, \boldsymbol{\omega}_M^\top, \boldsymbol{\Theta}_G^\top, \zeta, \xi]^\top$, and the dynamics are control-affine, $\dot{\hat{\mathbf{x}}} = f(\hat{\mathbf{x}}) + \sum_{i=1}^7 g_i(\hat{\mathbf{x}}) \hat{u}_i$, with $\hat{\mathbf{u}}_{MC} = [\ddot{T}, \tau_\phi, \tau_\theta, \tau_\psi, \dot{\phi}_G, \dot{\theta}_G, \dot{\psi}_G]^\top$ and

$$f = \begin{bmatrix} \mathbf{v} \\ \frac{1}{2} \mathbf{G}(\mathbf{q}_M) \boldsymbol{\omega}_M \\ \frac{\zeta}{m_M} \mathbf{b}_3 - g \mathbf{e}_3 \\ -\mathbf{J}_M^{-1} \boldsymbol{\omega}_M \times \mathbf{J}_M \boldsymbol{\omega}_M \\ \mathbf{0}_3, \xi, 0 \end{bmatrix}. \quad (37)$$

The only nonzero input channels are $g_1 = \partial/\partial \xi$ (thrust jerk), $[g_2, g_3, g_4] = \mathbf{J}_M^{-1}$ entering $\boldsymbol{\omega}_M$, and $[g_5, g_6, g_7] = \mathbf{I}_3$ entering $\boldsymbol{\Theta}_G$.

B. Relative degrees by repeated Lie differentiation

The outputs are $\mathbf{y}_O = [\mathbf{p}_M^\top, q_3, \boldsymbol{\Theta}_G^\top]^\top$.

Position. Differentiating along f ,

$$\begin{aligned}\dot{\mathbf{p}}_M &= \mathbf{v}, & \ddot{\mathbf{p}}_M &= \frac{\zeta}{m_M} \mathbf{b}_3 - g\mathbf{e}_3, \\ \mathbf{p}_M^{(3)} &= \frac{1}{m_M} \left(\xi \mathbf{b}_3 + \zeta \dot{\mathbf{b}}_3 \right), \\ \mathbf{p}_M^{(4)} &= \frac{1}{m_M} \left(\ddot{T} \mathbf{b}_3 + 2\xi \dot{\mathbf{b}}_3 + \zeta \ddot{\mathbf{b}}_3 \right),\end{aligned}\quad (38)$$

with $\dot{\mathbf{b}}_3 = \boldsymbol{\omega}_w \times \mathbf{b}_3$ and $\ddot{\mathbf{b}}_3 = \dot{\boldsymbol{\omega}}_w \times \mathbf{b}_3 + \boldsymbol{\omega}_w \times (\boldsymbol{\omega}_w \times \mathbf{b}_3)$. The inputs $\ddot{T}$ (through ξ) and τ_M (through $\dot{\boldsymbol{\omega}}_M$ in $\ddot{\mathbf{b}}_3$) first appear in $\mathbf{p}_M^{(4)}$, so $r_{1,2,3} = 4$. The double integrator is exactly what delays the thrust to the fourth derivative; without it $\mathbf{b}_3$ multiplies T already at the second derivative and Δ is singular.

Multicopter yaw. From $\dot{\mathbf{q}}_M = \frac{1}{2} \mathbf{G}(\mathbf{q}_M) \boldsymbol{\omega}_M$ one has $\dot{q}_3 = \frac{1}{2} \boldsymbol{\gamma}^\top \boldsymbol{\omega}_M$ with $\boldsymbol{\gamma}^\top = [-q_2, q_1, q_0]$, whose derivative contains $\dot{\boldsymbol{\omega}}_M = \mathbf{J}_M^{-1} \tau_M + \dots$; hence $r_4 = 2$.

Gimbal. $\dot{\boldsymbol{\Theta}}_G = \mathbf{u}_G$ directly, so $r_{5,6,7} = 1$.

Thus $\hat{\mathbf{r}} = [4, 4, 4, 2, 1, 1, 1]$ and $\sum_i \hat{r}_i = 17 = \hat{N}$: the extended system is exactly input–output linearizable with no zero dynamics.

C. Decoupling matrix, drift, and inverse

Writing the highest derivatives as $[\mathbf{p}_M^{(4)\top}, \ddot{q}_3, \dot{\boldsymbol{\Theta}}_G^\top]^\top = \mathbf{b}(\hat{\mathbf{x}}) + \Delta(\hat{\mathbf{x}}) \hat{\mathbf{u}}_{MC}$ and using $\dot{\boldsymbol{\omega}}_w|_{\text{input}} = \mathbf{R}(\mathbf{q}_M) \mathbf{J}_M^{-1} \tau_M$ in (38), with the torque-to-position coupling block $\mathbf{M}_\tau = -\frac{\zeta}{m_M} [\mathbf{b}_3]_\times \mathbf{R} \mathbf{J}_M^{-1}$,

$$\Delta = \begin{bmatrix} \frac{1}{m_M} \mathbf{b}_3 & \mathbf{M}_\tau & \mathbf{0} \\ \mathbf{0} & \frac{1}{2} \boldsymbol{\gamma}^\top \mathbf{J}_M^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I}_3 \end{bmatrix}, \quad (39)$$

where the block columns are the thrust-jerk, torque, and gimbal-rate channels, and the block rows are the position, yaw, and gimbal outputs. The gimbal block $\mathbf{I}_3$ factors out, leaving a four-dimensional thrust–attitude block whose determinant satisfies

$$\det \Delta \propto \zeta q_0, \quad (40)$$

up to a nonzero factor depending on m_M and $\det \mathbf{J}_M$. Indeed, $[\mathbf{b}_3]_\times$ projects the torque channel onto the plane orthogonal to $\mathbf{b}_3$, which collapses as the thrust $\zeta \rightarrow 0$, while $\boldsymbol{\gamma}^\top$ degenerates as $q_0 \rightarrow 0$ (the multicopter fully inverted). Hence Δ is nonsingular whenever $\zeta \neq 0$ and $q_0 \neq 0$, and the law is

$$\hat{\mathbf{u}}_{MC} = \alpha(\hat{\mathbf{x}}) + \beta(\hat{\mathbf{x}}) \mathbf{v}_{MC}, \quad \alpha = -\Delta^{-1} \mathbf{b}, \quad \beta = \Delta^{-1}. \quad (41)$$

The symbolically generated α and β carry exactly the factors $1/\zeta$ and $1/q_0$ predicted by (40); the latter is the inverted-flight representation singularity that the damped inverse of Section IV-A regularizes by flooring $|q_0|$.

References

- [1] S. Shah, D. Dey, C. Lovett, and A. Kapoor, “AirSim: High-fidelity cross-platform simulator for autonomous vehicles,” in *Field and Service Robotics (FSR)*, 2017.
- [2] A. Amer, O. Álvarez-Tuñón, H. İ. Uğurlu, J. L. F. Sejersten, Y. Brodskiy, and E. Kayacan, “Unav-sim: A visually realistic underwater robotics simulator and synthetic data-generation framework,” in *2023 21st International Conference on Advanced Robotics (ICAR)*, 2023, pp. 570–576.
- [3] Y. Gao, J. Lin, and F. Wu, “The reality gap: A survey of sim-to-real transfer in robotics,” in *2020 IEEE International Conference on Robotics and Automation (ICRA)*, 2020, pp. 5678–5684.
- [4] E. Aljalbout, J. Xing, A. Romero, I. Akinola, C. R. Garrett, E. Heiden, A. Gupta, T. Hermans, Y. Narang, D. Fox et al., “The reality gap in robotics: Challenges, solutions, and best practices,” *arXiv preprint arXiv:2510.20808*, 2025.
- [5] O. Stroosma, M. Van Paassen, and M. Mulder, “Using the simona research simulator for human-machine interaction research,” in *AIAA modeling and simulation technologies conference and exhibit*, 2003, p. 5525.
- [6] S. Edvani, E. va, S. Bettendorf, S. Edvani, E. va, and S. Bettendorf, “Simona-a reconfigurable and versatile research facility,” in *Modeling and Simulation Technologies Conference*, 1997, p. 3809.
- [7] M. M. van Paassen, M. Mulder, and J. C. van der Vaart, “The SIMONA research simulator: A reconfigurable simulation environment for aerospace research,” in *AIAA Modeling and Simulation Technologies Conference and Exhibit*, ser. AIAA 2006-6182, 2006.
- [8] J. Smith and B. Anderson, “A review of dynamic emulation for validation of autonomous systems,” *Journal of Field Robotics*, vol. 39, no. 2, pp. 145–168, 2022.
- [9] X. Yuan, J. Xu, and S. Li, “Design and simulation verification of model predictive attitude control based on feedback linearization for quadrotor uav,” *Applied Sciences*, vol. 15, no. 9, p. 5218, 2025.
- [10] H. Wu, Z. Wang, Z. Zhou, and R. Wang, “Modeling and simulation for multi-rotor fixed-wing uav based on multibody dynamics,” *Journal of Northwestern Polytechnical University*, vol. 37, no. 5, pp. 928–934, 2019.
- [11] X. Wang, Y. Li, and Y. Han, “Similarity between two dynamical systems,” 2023. [Online]. Available: <https://arxiv.org/abs/2310.03383>
- [12] K. Xiao, Y. Meng, X. Dai, H. Zhang, and Q. Quan, “A lifting wing fixed on multirotor uavs for long flight ranges,” in *2021 International Conference on Unmanned Aircraft Systems (ICUAS)*. IEEE, 2021, pp. 1605–1610.
- [13] Q. Quan, S. Wang, and W. Gao, “Lifting-wing quadcopter modeling and unified control,” *Journal of Guidance, Control, and Dynamics*, vol. 48, no. 3, pp. 689–699, 2025.
- [14] A. Amer, M. Mehndiratta, J. le Fevre Sejersten, H. X. Pham, and E. Kayacan, “Visual tracking nonlinear model predictive control method for autonomous wind turbine inspection,” in *2023 21st International Conference on Advanced Robotics (ICAR)*. IEEE, 2023, pp. 431–438.
- [15] A. Amer, M. Mehndiratta, Y. Brodskiy, and E. Kayacan, “Empowering autonomous underwater vehicles using learning-based model predictive control with dynamic forgetting gaussian processes,” *IEEE Transactions on Control Systems Technology*, 2025.
- [16] K. Alexis, C. Papachristos, R. Siegwart, and A. Tzes, “Model predictive control for UAV trajectory tracking in complex environments,” in *2016 European Control Conference (ECC)*, 2016, pp. 2200–2207.
- [17] A. Isidori, *Nonlinear Control Systems*, 3rd ed., ser. Communications and Control Engineering. London: Springer-Verlag, 1995.
- [18] A. Isidori, A. J. Krener, C. Gori-Giorgi, and S. Monaco, “Nonlinear decoupling via feedback: A differential geometric approach,” *IEEE Transactions on Automatic Control*, vol. 26, no. 2, pp. 331–345, 1981.
- [19] A. Isidori, C. H. Moog, and A. D. Luca, “A sufficient condition for full linearization via dynamic state feedback,” in *1986 25th IEEE Conference on Decision and Control*, 1986, pp. 203–208.

- [20] V. Mistler, A. Benallegue, and N. M'Sirdi, "Exact linearization and noninteracting control of a 4 rotors helicopter via dynamic feedback," in Proceedings 10th IEEE International Workshop on Robot and Human Interactive Communication. ROMAN 2001 (Cat. No.01TH8591), 2001, pp. 586–593.
- [21] M. Fliess, J. Lévine, P. Martin, and P. Rouchon, "Flatness and defect of non-linear systems: Introductory theory and examples," *International Journal of Control*, vol. 61, no. 6, pp. 1327–1361, 1995.
- [22] D. Mellinger and V. Kumar, "Minimum snap trajectory generation and control for quadrotors," in 2011 IEEE International Conference on Robotics and Automation (ICRA), 2011, pp. 2520–2525.
- [23] M. Faessler, A. Franchi, and D. Scaramuzza, "Differential flatness of quadrotor dynamics subject to rotor drag for accurate tracking of high-speed trajectories," *IEEE Robotics and Automation Letters*, vol. 3, no. 2, pp. 620–626, 2018.
- [24] T. Lee, M. Leok, and N. H. McClamroch, "Geometric tracking control of a quadrotor UAV on $SE(3)$," in 49th IEEE Conference on Decision and Control (CDC), 2010, pp. 5420–5425.
- [25] A. Isidori and C. I. Byrnes, "Output regulation of nonlinear systems," *IEEE Transactions on Automatic Control*, vol. 35, no. 2, pp. 131–140, 1990.
- [26] A. Sarabakha, "Data-driven identification of observed relative degrees for nonlinear systems," in 2025 European Control Conference (ECC). IEEE, 2025, pp. 813–818.
- [27] D. C. Eyre, "Zivko edge 540 — aircraft model," 2010, "Encyclopedia of Aircraft" web article, wingspan = 7.62 m, wing area = 9.10 m², rate of roll = 420°/s, etc. [Online]. Available: <https://aeropedia.com.au/content/zivko-edge-540/>